

Optimization

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Optimality conditions in several variables

Source

This section is adapted from the multivariate calculus and optimality-condition parts of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source states the second-order Taylor approximation and the optimality conditions without [proof](#); here they are proved from the one-variable Taylor theorem, and the approximation is made exact by evaluating the Hessian at a point between $\tilde{z}$ and $\tilde{z} + \tilde{h}$. The source calls points with zero gradient critical points; here they are stationary points, because the [calculus notes](#) give “critical point” a wider meaning. These parts of the source are not part of this section:

- its one-variable Taylor example and its plots
- matrix calculus
- convexity
- Lagrange multipliers
- Newton’s method and gradient descent

A local minimizer of a differentiable function on an open interval is a [flat point](#), and, at a flat point, a positive [second derivative](#) guarantees a strict local minimum. This section extends both facts to a function f of a $p \times 1$ vector $\tilde{x}$: the gradient $\frac{\partial}{\partial \tilde{x}} f(\tilde{x})$ ([gradient](#)) takes the place of the first derivative, and a $p \times p$ matrix of second partial derivatives takes the place of the second; both facts are proved in this section, for every $p \geq 1$. Throughout, $\frac{\partial}{\partial \tilde{x}} f(\tilde{z})$ means the gradient evaluated at $\tilde{x} = \tilde{z}$.

Theorem 0.1 (A minimizer has derivative zero). *Let g be a [real-valued function](#) on an [open interval](#) containing c , and let g be [differentiable](#) at c . If c is a [local minimizer](#) of g , then $g'(c) = 0$.*

i Proof

Proof. Take $\delta > 0$ as in the definition of a local minimizer, shrunk if needed so that $(c - \delta, c + \delta)$ lies inside the interval; then $g(c) \leq g(c + h)$ whenever $|h| < \delta$. For $0 < h < \delta$, the difference quotient is

$$\frac{g(c + h) - g(c)}{h} \geq 0,$$

because its numerator is at least 0 and its denominator is positive. As $h \rightarrow 0$ from the right, these quotients tend to $g'(c)$ (the two-sided limit exists, so the [one-sided limit](#) equals it), and a limit of numbers that are all at least 0 is at least 0; so $g'(c) \geq 0$. For $-\delta < h < 0$, the numerator is still at least 0 but the denominator is negative, so the quotient is at most 0, and in the same way, with $h \rightarrow 0$ from the left, $g'(c) \leq 0$. Together, $g'(c) = 0$. $\square$

Example 0.1 (Applying the test, and its limits).

- $g(x) = (x - 2)^2$ has its minimizer at $c = 2$, and $g'(2) = 2(2 - 2) = 0$.
- The [converse](#) fails: $g(x) = x^3$ has $g'(0) = 3 \cdot 0^2 = 0$, but 0 is not a local minimizer, since $g(-h) = -h^3 < 0 = g(0)$ for every $h > 0$.
- Differentiability is needed: $g(x) = |x|$ has its minimizer at 0, but the difference quotient $|h|/h$ is 1 for $h > 0$ and -1 for $h < 0$, so it has no limit and $g'(0)$ does not exist.

Definition 0.1 (Stationary point). Let f be a scalar-valued function of a $p \times 1$ vector $\tilde{x}$. A point $\tilde{z}$ at which the gradient exists and

$$\frac{\partial}{\partial \tilde{x}} f(\tilde{z}) = \tilde{0}_{p \times 1}$$

is a **stationary point** of f . For $p = 1$, this is a [flat point](#).

Example 0.2 (A stationary point, and a point that is not one). Let $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$. Its gradient is $(2x_1 - 2, 2x_2)^\top$, which is $\tilde{0}_{2 \times 1}$ exactly when $x_1 = 1$ and $x_2 = 0$. So $(1, 0)^\top$ is its only stationary point, and $\tilde{0}$ is not one: the gradient there is $(-2, 0)^\top$.

Theorem 0.2 (First-order necessary condition). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$. If $\tilde{x}^*$ is a [local minimizer](#) of f and the gradient of f exists at $\tilde{x}^*$, then $\tilde{x}^*$ is a stationary point of f (Definition 0.1).

i Proof

Proof. Take $\delta > 0$ as in the definition of a local minimizer. Fix $i \in \{1, \dots, p\}$, let $\tilde{e}_i$ be the vector with 1 in entry i and 0 elsewhere, and let $g_i(t) \stackrel{\text{def}}{=} f(\tilde{x}^* + t\tilde{e}_i)$ for $t \in \mathbb{R}$. For $|t| < \delta$, $\|(\tilde{x}^* + t\tilde{e}_i) - \tilde{x}^*\| = |t|\|\tilde{e}_i\| = |t| < \delta$, so $g_i(0) = f(\tilde{x}^*) \leq f(\tilde{x}^* + t\tilde{e}_i) = g_i(t)$: 0 is a local minimizer of g_i . Moving $\tilde{x}$ from $\tilde{x}^*$ along $\tilde{e}_i$ changes only x_i , so $g'_i(0)$ is the partial derivative $\frac{\partial}{\partial x_i} f(\tilde{x}^*)$, which exists because the gradient does. By Theorem 0.1, $\frac{\partial}{\partial x_i} f(\tilde{x}^*) = g'_i(0) = 0$. The equation $g'_i(0) = 0$ holds for every i , so every entry of the gradient at $\tilde{x}^*$ is 0 ([vector derivative](#)). $\square$

Example 0.3 (Using the condition to locate, and to rule out, minimizers).

- For $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ of Example 0.2, any local minimizer must be the stationary point $(1, 0)^\top$. It is one: completing the square, $f(\tilde{x}) = (x_1 - 1)^2 + x_2^2 - 1 \geq -1 = f(1, 0)$.
- For $f(\tilde{x}) = e^{2x_1+x_2} - x_1$ of [Example](#), the second entry of the gradient is $e^{2x_1+x_2} > 0$, so f has no stationary point, and so no local minimizer.
- The converse fails: $f(\tilde{x}) = x_1^2 - x_2^2$ has gradient $(2x_1, -2x_2)^\top$, so $\tilde{0}$ is a stationary point, but $f(0, t) = -t^2 < 0 = f(\tilde{0})$ for every $t \neq 0$, so $\tilde{0}$ is not a local minimizer.

Theorem 0.3 (Taylor's theorem with a second-order remainder). *Let $a < b$, and let g be a real-valued function on $[a, b]$ whose derivative g' is [continuous](#) on $[a, b]$ and whose [second derivative](#) g'' exists at every point of (a, b) . Then there is a $\tau \in (a, b)$ with*

$$g(b) = g(a) + g'(a)(b - a) + \frac{1}{2}g''(\tau)(b - a)^2.$$

This theorem is the case $n = 2$ of (Rudin 1976, Theorem 5.15); its proof uses the mean value theorem, which these notes do not develop.

Example 0.4 (The remainder point for e^x). Take $g(x) = e^x$ on $[0, 1]$, so $g' = g'' = g$. Theorem 0.3 says $e = 1 + 1 + \frac{1}{2}e^\tau$ for some $\tau \in (0, 1)$. Solving, $e^\tau = 2(e - 2) \approx 2 \times 0.71828 = 1.43656$, so $\tau = \log 1.43656 \approx 0.362$ ([natural logarithm](#)), which is in $(0, 1)$.

Theorem 0.4 (Second-order Taylor theorem in several variables). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$ ([continuity](#)), and let $\tilde{z}, \tilde{h} \in \mathbb{R}^p$. Then there is a $\tau \in (0, 1)$ with*

$$f(\tilde{z} + \tilde{h}) = f(\tilde{z}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{z}) \right)^\top \tilde{h} + \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{z} + \tau \tilde{h}) \tilde{h}.$$

i Proof

Proof. Let $g(t) \stackrel{\text{def}}{=} f(\tilde{z} + t\tilde{h})$ for $t \in \mathbb{R}$, the values of f along the line through $\tilde{z}$ in the direction $\tilde{h}$. The **inner function** $\tilde{y}(t) = \tilde{z} + t\tilde{h}$ has entries $y_j = z_j + th_j$, so $\frac{\partial}{\partial t}y_j = h_j$, and its derivative (**vector-valued derivative**) is the $1 \times p$ matrix $\tilde{h}^\top$. By the vector chain rule (**vector chain rule**, with its input t of length 1, its inner function $\tilde{y}(t) = \tilde{z} + t\tilde{h}$, and its outer function f),

$$\begin{aligned} g'(t) &= \tilde{h}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{z} + t\tilde{h}) && \text{(vector chain rule)} \\ &= \sum_{j=1}^p h_j \frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}). && \text{(matrix product)} \end{aligned}$$

The second step writes out the **matrix product**. g' is continuous (**continuity**): $t \mapsto \tilde{z} + t\tilde{h}$ is continuous, since $\|(\tilde{z} + s\tilde{h}) - (\tilde{z} + t\tilde{h})\| = |s - t|\|\tilde{h}\|$, each $\frac{\partial}{\partial x_j} f$ is continuous by assumption, and compositions, sums and constant multiples of continuous functions are continuous. Each $\frac{\partial}{\partial x_j} f$ has continuous partial derivatives, the second partial derivatives of f , so the same chain-rule computation applies to it: $\frac{\partial}{\partial t} \left[\frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}) \right] = \sum_{i=1}^p h_i \frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}) \right)$. So

$$\begin{aligned} g''(t) &= \sum_{j=1}^p h_j \sum_{i=1}^p h_i \frac{\partial}{\partial x_i} \left(\frac{\partial}{\partial x_j} f(\tilde{z} + t\tilde{h}) \right) && \text{(differentiate each term of } g'(t)) \\ &= \sum_{i=1}^p \sum_{j=1}^p h_i [\mathbf{H}_f(\tilde{z} + t\tilde{h})]_{ij} h_j && \text{(definition of the Hessian; reorder the finite sums)} \\ &= \tilde{h}^\top \mathbf{H}_f(\tilde{z} + t\tilde{h}) \tilde{h}. && \text{(matrix product)} \end{aligned}$$

The last step is again the **matrix product**. Now apply Theorem 0.3 to g on $[0, 1]$: there is a $\tau \in (0, 1)$ with

$$\begin{aligned} f(\tilde{z} + \tilde{h}) &= g(1) && \text{(definition of } g) \\ &= g(0) + g'(0)(1 - 0) + \frac{1}{2} g''(\tau)(1 - 0)^2 && \text{(Theorem 0.3, with } a = 0, b = 1) \\ &= g(0) + g'(0) + \frac{1}{2} g''(\tau) && (1 - 0 = 1) \\ &= f(\tilde{z}) + \tilde{h}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{z}) + \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{z} + \tau\tilde{h}) \tilde{h} && \text{(substitute } g(0), g'(0), g''(\tau)) \\ &= f(\tilde{z}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{z}) \right)^\top \tilde{h} + \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{z} + \tau\tilde{h}) \tilde{h}. && \text{(a } 1 \times 1 \text{ matrix equals its transpose)} \end{aligned}$$

□

Example 0.5 (For a quadratic form the expansion is exact). Let $f(\tilde{x}) = \tilde{x}^\top \mathbf{S} \tilde{x}$ with $\mathbf{S}$ symmetric and constant. Its gradient is $2\mathbf{S}\tilde{x}$ (derivative of a quadratic form) and its Hessian is $2\mathbf{S}$ at every point (Hessian of a quadratic form), so whatever τ is, Theorem 0.4 reads

$$f(\tilde{z} + \tilde{h}) = \tilde{z}^\top \mathbf{S} \tilde{z} + (2\mathbf{S}\tilde{z})^\top \tilde{h} + \frac{1}{2} \tilde{h}^\top (2\mathbf{S}) \tilde{h} = \tilde{z}^\top \mathbf{S} \tilde{z} + 2 \tilde{z}^\top \mathbf{S} \tilde{h} + \tilde{h}^\top \mathbf{S} \tilde{h},$$

using $(2\mathbf{S}\tilde{z})^\top = 2 \tilde{z}^\top \mathbf{S}^\top = 2 \tilde{z}^\top \mathbf{S}$. Multiplying out directly gives the same:

$$\begin{aligned} (\tilde{z} + \tilde{h})^\top \mathbf{S} (\tilde{z} + \tilde{h}) &= \tilde{z}^\top \mathbf{S} \tilde{z} + \tilde{z}^\top \mathbf{S} \tilde{h} + \tilde{h}^\top \mathbf{S} \tilde{z} + \tilde{h}^\top \mathbf{S} \tilde{h} \quad (\text{distribute}) \\ &= \tilde{z}^\top \mathbf{S} \tilde{z} + 2 \tilde{z}^\top \mathbf{S} \tilde{h} + \tilde{h}^\top \mathbf{S} \tilde{h}. \quad (\tilde{h}^\top \mathbf{S} \tilde{z} = (\tilde{h}^\top \mathbf{S} \tilde{z})^\top = \tilde{z}^\top \mathbf{S} \tilde{h}) \end{aligned}$$

Definition 0.2 (Strict local minimizer). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$. A point $\tilde{x}^*$ is a **strict local minimizer** of f if there is a number $\delta > 0$ such that $f(\tilde{x}^*) < f(\tilde{x})$ for every $\tilde{x} \neq \tilde{x}^*$ with $\|\tilde{x} - \tilde{x}^*\| < \delta$.

Example 0.6 (Strict and non-strict local minimizers).

- $f(x) = (x - 2)^2$ has $f(2) = 0 < (x - 2)^2 = f(x)$ for every $x \neq 2$, so 2 is a strict local minimizer (any $\delta > 0$ works).
- A constant function $f(\tilde{x}) = 0$ has $f(\tilde{x}^*) \leq f(\tilde{x})$ for all $\tilde{x}^*$ and $\tilde{x}$, so every point is a **local minimizer**, but no point is a strict one: $f(\tilde{x}^*) < f(\tilde{x})$ never holds.

Lemma 0.1 (A quadratic form is at least the smallest eigenvalue times the squared length). Let $\mathbf{A}$ be a symmetric $p \times p$ matrix, and let $\lambda_{\min}$ be the smallest of its eigenvalues (eigendecomposition). Then $\tilde{h}^\top \mathbf{A} \tilde{h} \geq \lambda_{\min} \|\tilde{h}\|^2$ for every $\tilde{h} \in \mathbb{R}^p$.

i Proof

Proof. Write $\mathbf{A} = \mathbf{Q}\mathbf{Q}^\top$ with $\mathbf{Q}$ orthogonal and $\mathbf{Q} = \text{diag}(\lambda_1, \dots, \lambda_p)$ (spectral theorem), and let $\tilde{y} \stackrel{\text{def}}{=} \mathbf{Q}^\top \tilde{h}$. Then

$$\begin{aligned}
\tilde{h}^\top \mathbf{A} \tilde{h} &= \tilde{h}^\top \mathbf{Q} \mathbf{Q}^\top \tilde{h} && \text{(substitute the eigendecomposition)} \\
&= \tilde{y}^\top \tilde{y} && (\tilde{h}^\top \mathbf{Q} = (\mathbf{Q}^\top \tilde{h})^\top, \text{ transpose of a product}) \\
&= \sum_{i=1}^p \lambda_i y_i^2 && (\text{ is diagonal}) \\
&\geq \sum_{i=1}^p \lambda_{\min} y_i^2 && (\lambda_i \geq \lambda_{\min} \text{ and } y_i^2 \geq 0) \\
&= \lambda_{\min} \|\tilde{y}\|^2 && (\text{the squared length is the sum of squares}) \\
&= \lambda_{\min} \|\tilde{h}\|^2. && (\text{orthogonal matrices preserve length})
\end{aligned}$$

The second step is the [transpose of a product](#). The last step applies [orthogonal matrices preserve length](#) to $\mathbf{Q}^\top$, which is orthogonal because $\mathbf{Q}^\top \mathbf{Q} = \mathbf{Q} \mathbf{Q}^\top = \mathbf{I}_p$ ([orthogonal matrix](#)). $\square$

Example 0.7 (Checking the bound). $\mathbf{A} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ has eigenvalues 3 and 1 ([eigenvalue example](#)), so $\lambda_{\min} = 1$. At $\tilde{h} = (1, 0)^\top$, $\tilde{h}^\top \mathbf{A} \tilde{h} = 2 \geq 1 \cdot 1$. At $\tilde{h} = (1, -1)^\top$, $\tilde{h}^\top \mathbf{A} \tilde{h} = 2 - 1 - 1 + 2 = 2 = 1 \cdot 2$: the bound holds with equality, since this $\tilde{h}$ is an eigenvector for $\lambda_{\min}$.

Lemma 0.2 (A quadratic form with small entries is small). *Let $\mathbf{E}$ be a $p \times p$ matrix whose entries all satisfy $|e_{ij}| \leq m$. Then $|\tilde{h}^\top \mathbf{E} \tilde{h}| \leq p m \|\tilde{h}\|^2$ for every $\tilde{h} \in \mathbb{R}^p$.*

i Proof

Proof. Let $\tilde{a} \stackrel{\text{def}}{=} (|h_1|, \dots, |h_p|)^\top$ and $\tilde{1} \stackrel{\text{def}}{=} (1, \dots, 1)^\top$, both in $\mathbb{R}^p$. Then

$$\begin{aligned}
|\tilde{h}^\top \mathbf{E} \tilde{h}| &= \left| \sum_{i=1}^p \sum_{j=1}^p h_i e_{ij} h_j \right| && \text{(matrix product)} \\
&\leq \sum_{i=1}^p \sum_{j=1}^p |h_i| |e_{ij}| |h_j| && \text{(triangle inequality for numbers)} \\
&\leq m \sum_{i=1}^p \sum_{j=1}^p |h_i| |h_j| && (|e_{ij}| \leq m) \\
&= m \left(\sum_{i=1}^p |h_i| \right)^2 && \text{(the double sum factors)} \\
&= m (\tilde{\mathbf{1}} \cdot \tilde{a})^2 && \text{(dot product)} \\
&\leq m \|\tilde{\mathbf{1}}\|^2 \|\tilde{a}\|^2 && \text{(Cauchy-Schwarz, squared)} \\
&= m p \|\tilde{h}\|^2. && (\|\tilde{\mathbf{1}}\|^2 = p \text{ and } \|\tilde{a}\|^2 = \sum_i h_i^2 = \|\tilde{h}\|^2)
\end{aligned}$$

The steps use the [matrix product](#), the [dot product](#) and the [Cauchy-Schwarz inequality](#). $\square$

Example 0.8 (Checking the bound). Let $\mathbf{E} = \begin{bmatrix} 0.1 & -0.2 \\ -0.2 & 0.1 \end{bmatrix}$, so $m = 0.2$ works, and $\tilde{h} = (1, 1)^\top$, so $\|\tilde{h}\|^2 = 2$. Then $\tilde{h}^\top \mathbf{E} \tilde{h} = 0.1 - 0.2 - 0.2 + 0.1 = -0.2$, and $|-0.2| = 0.2 \leq 2 \cdot 0.2 \cdot 2 = 0.8$.

Theorem 0.5 (Second-order sufficient condition). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$ ([continuity](#)). If $\tilde{x}^*$ is a stationary point of f (Definition 0.1) and $\mathbf{H}_f(\tilde{x}^*)$ is [positive definite](#), then $\tilde{x}^*$ is a strict local minimizer of f (Definition 0.2).*

i Proof

Proof. $\mathbf{H}_f(\tilde{x}^*)$ is positive definite, and so symmetric, so its smallest eigenvalue λ is positive ([definiteness and eigenvalues](#)). Each of the p^2 entries of $\mathbf{H}_f$ is continuous at $\tilde{x}^*$ ([continuity](#)): for $\varepsilon = \lambda/(2p)$ there is a $\delta_{ij} > 0$ with $|\mathbf{H}_f(\tilde{y})_{ij} - \mathbf{H}_f(\tilde{x}^*)_{ij}| < \lambda/(2p)$ whenever $\|\tilde{y} - \tilde{x}^*\| < \delta_{ij}$. Let δ be the smallest of these p^2 numbers.

Now let $0 < \|\tilde{h}\| < \delta$. By Theorem 0.4 there is a $\tau \in (0, 1)$ for which the first step of the next display holds; let $\tilde{y} \stackrel{\text{def}}{=} \tilde{x}^* + \tau \tilde{h}$ and $\mathbf{E} \stackrel{\text{def}}{=} \mathbf{H}_f(\tilde{y}) - \mathbf{H}_f(\tilde{x}^*)$. Since $\|\tilde{y} - \tilde{x}^*\| = \tau \|\tilde{h}\| < \delta$, every entry of $\mathbf{E}$ has absolute value less than $\lambda/(2p)$, so Lemma 0.2 applies with $m = \lambda/(2p)$:

$$\begin{aligned}
f(\tilde{x}^* + \tilde{h}) - f(\tilde{x}^*) &= \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{x}^*) \right)^\top \tilde{h} + \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{y}) \tilde{h} && \text{(Theorem 0.4)} \\
&= \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{y}) \tilde{h} && (\tilde{x}^* \text{ is stationary}) \\
&= \frac{1}{2} \tilde{h}^\top \mathbf{H}_f(\tilde{x}^*) \tilde{h} + \frac{1}{2} \tilde{h}^\top \mathbf{E} \tilde{h} && (\mathbf{H}_f(\tilde{y}) = \mathbf{H}_f(\tilde{x}^*) + \mathbf{E}; \text{ distribute}) \\
&\geq \frac{1}{2} \lambda \|\tilde{h}\|^2 + \frac{1}{2} \tilde{h}^\top \mathbf{E} \tilde{h} && \text{(Lemma 0.1)} \\
&\geq \frac{1}{2} \lambda \|\tilde{h}\|^2 - \frac{1}{2} |\tilde{h}^\top \mathbf{E} \tilde{h}| && \text{(a number is at least minus its absolute value)} \\
&\geq \frac{1}{2} \lambda \|\tilde{h}\|^2 - \frac{1}{2} p \frac{\lambda}{2p} \|\tilde{h}\|^2 && \text{(Lemma 0.2, } m = \lambda/(2p)) \\
&= \frac{\lambda}{4} \|\tilde{h}\|^2 && \text{(arithmetic)} \\
&> 0. && (\lambda > 0 \text{ and } \tilde{h} \neq \tilde{0})
\end{aligned}$$

So $f(\tilde{x}^*) < f(\tilde{x})$ whenever $0 < \|\tilde{x} - \tilde{x}^*\| < \delta$ (take $\tilde{h} = \tilde{x} - \tilde{x}^*$). $\square$

Example 0.9 (Classifying stationary points).

- $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ has the stationary point $(1, 0)^\top$ (Example 0.2), and its Hessian is $\mathbf{H}_f(\tilde{x}) = \begin{bmatrix} 2 & 0 \\ 0 & 2 \end{bmatrix} = 2\mathbf{I}_2$, which is positive definite, so $(1, 0)^\top$ is a strict local minimizer.
- $f(\tilde{x}) = x_1^2 - x_2^2$ has the stationary point $\tilde{0}$ and Hessian $\begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$, which is not positive definite ($(0, 1)^\top$ gives -2), and indeed $\tilde{0}$ is not a local minimizer (Example 0.3).

Example 0.10 (The condition is sufficient, not necessary). $f(\tilde{x}) = x_1^2 + x_2^4$ and $g(\tilde{x}) = x_1^2 - x_2^4$ both have gradient $\tilde{0}$ at $\tilde{0}$ and the same Hessian there, $\begin{bmatrix} 2 & 0 \\ 0 & 0 \end{bmatrix}$, since $\frac{\partial}{\partial x_2} \left(\frac{\partial}{\partial x_2} (\pm x_2^4) \right) = \pm 12x_2^2 = 0$ at $x_2 = 0$. That Hessian is not positive definite ($(0, 1)^\top$ gives 0), so Theorem 0.5 says nothing about either function. In fact $\tilde{0}$ is a strict local minimizer of f , since $f(\tilde{x}) > 0 = f(\tilde{0})$ for $\tilde{x} \neq \tilde{0}$, but not a local minimizer of g , since $g(0, t) = -t^4 < 0 = g(\tilde{0})$ for $t \neq 0$.

Convexity in several variables

Source

This section is adapted from the convexity part of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source states the first-order test (in one direction) and the second-derivative test for convexity without proof; here both directions of the first-order test are proved, and the second-derivative test is proved from the Taylor theorem of Section . The source defines both concave and strictly concave as “ $-f$ is strictly convex”; here f is concave when $-f$ is convex. These parts of the source are not part of this section:

- its catalogs of convex sets and convex functions, including the positive semidefinite matrices and $-\log \det$
- intersections of infinitely many convex sets, and suprema of infinitely many convex functions
- its examples, exercises and figures, including the multivariate normal maximum likelihood example

Definition 0.3 (Convex set). A set $K \subseteq \mathbb{R}^p$ is **convex** if for all $\tilde{x}, \tilde{y} \in K$ and all $t \in [0, 1]$, the point $t\tilde{x} + (1-t)\tilde{y}$ is in K : K contains the whole line segment between any two of its points.

Example 0.11 (Convex sets).

- $\mathbb{R}^p$ is convex, and so is any set with at most one point.
- The closed ball ([ball](#)) $B = \{\tilde{x} : \|\tilde{x}\| \leq r\}$ is convex: for $\tilde{x}, \tilde{y} \in B$ and $t \in [0, 1]$,

$$\begin{aligned} \|t\tilde{x} + (1-t)\tilde{y}\| &\leq \|t\tilde{x}\| + \|(1-t)\tilde{y}\| && \text{(triangle inequality)} \\ &= t\|\tilde{x}\| + (1-t)\|\tilde{y}\| && \text{(norm of a multiple; } t, 1-t \geq 0) \\ &\leq tr + (1-t)r && (\tilde{x}, \tilde{y} \in B) \\ &= r. && \text{(arithmetic)} \end{aligned}$$

The first step is the [triangle inequality](#), and the second uses [the norm of a multiple](#).

Example 0.12 (A set that is not convex). $K = \{x \in \mathbb{R} : |x| \geq 1\}$ is not convex: -1 and 1 are in K , but with $t = \frac{1}{2}$ the point $\frac{1}{2}(-1) + \frac{1}{2}(1) = 0$ is not.

Definition 0.4 (Half-space). Let $\tilde{a} \in \mathbb{R}^p$ be a nonzero vector and c a real number. The set $\{\tilde{x} \in \mathbb{R}^p : \tilde{a}^\top \tilde{x} \geq c\}$ is a (closed) **half-space**: the points on one side of the boundary $\{\tilde{x} : \tilde{a}^\top \tilde{x} = c\}$, together with the boundary. In $\mathbb{R}^2$, a half-space is called a **half-plane**.

Example 0.13 (The right half-plane). With $\tilde{a} = (1, 0)^\top$ and $c = 0$, the half-plane $\{\tilde{x} \in \mathbb{R}^2 : x_1 \geq 0\}$ is the set of points on or to the right of the x_2 -axis. $(1, 5)^\top$ is in it, since $1 \geq 0$; $(0, -3)^\top$ is in it, on the boundary; and $(-1, 0)^\top$ is not, since $-1 < 0$.

Theorem 0.6 (An intersection of convex sets is convex). *If K_1 and K_2 are convex subsets of $\mathbb{R}^p$ (Definition 0.3), then $K_1 \cap K_2$ is convex.*

i Proof

Proof. Let $\tilde{x}, \tilde{y} \in K_1 \cap K_2$ and $t \in [0, 1]$. Then $\tilde{x}, \tilde{y} \in K_1$, so $t\tilde{x} + (1-t)\tilde{y} \in K_1$, since K_1 is convex; in the same way $t\tilde{x} + (1-t)\tilde{y} \in K_2$. So $t\tilde{x} + (1-t)\tilde{y} \in K_1 \cap K_2$. $\square$

Example 0.14 (Intersecting a ball with a half-plane). The closed unit ball (ball) in $\mathbb{R}^2$ is convex (Example 0.11), and so is the half-plane (Definition 0.4) $\{\tilde{x} : x_1 \geq 0\}$: if $x_1 \geq 0$ and $y_1 \geq 0$, then $tx_1 + (1-t)y_1 \geq 0$ for $t \in [0, 1]$. By Theorem 0.6, the closed half-disk $\{\tilde{x} : \|\tilde{x}\| \leq 1, x_1 \geq 0\}$ is convex. A union need not be convex: $\{x \in \mathbb{R} : x \leq -1\}$ and $\{x \in \mathbb{R} : x \geq 1\}$ are convex, but their union is the set of Example 0.12.

Definition 0.5 (Strictly convex and concave functions). A function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is **strictly convex** if $f(t\tilde{x} + (1-t)\tilde{y}) < tf(\tilde{x}) + (1-t)f(\tilde{y})$ for all $\tilde{x} \neq \tilde{y}$ in $\mathbb{R}^p$ and all $t \in (0, 1)$. It is **concave** if $-f$ is **convex**, and **strictly concave** if $-f$ is strictly convex.

Example 0.15 (Strictly convex, and convex but not strictly).

- $f(x) = x^2$ is strictly convex:

$$\begin{aligned} tx^2 + (1-t)y^2 - (tx + (1-t)y)^2 &= tx^2 + (1-t)y^2 - t^2x^2 - 2t(1-t)xy - (1-t)^2y^2 \\ &= t(1-t)x^2 - 2t(1-t)xy + t(1-t)y^2 \\ &= t(1-t)(x-y)^2, \end{aligned}$$

(expand the square)
($t - t^2 = t(1-t)$)
(factor)

which is positive when $x \neq y$ and $t \in (0, 1)$. At $t \in \{0, 1\}$ or $x = y$ it is 0, so x^2 is also convex.

- $f(x) = 2x + 1$ is convex but not strictly convex: $f(tx + (1-t)y) = 2tx + 2(1-t)y + 1 = tf(x) + (1-t)f(y)$ for every x, y, t , so the inequality holds, but never strictly. It is also concave.

Theorem 0.7 (The maximum of convex functions is convex). *If f_1 and f_2 are [convex](#) functions on $\mathbb{R}^p$, then $f(\tilde{x}) \stackrel{\text{def}}{=} \max\{f_1(\tilde{x}), f_2(\tilde{x})\}$ is convex.*

i Proof

Proof. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$ and $t \in [0, 1]$. For $i = 1, 2$,

$$\begin{aligned} f_i(t\tilde{x} + (1-t)\tilde{y}) &\leq tf_i(\tilde{x}) + (1-t)f_i(\tilde{y}) \quad (f_i \text{ is convex}) \\ &\leq tf(\tilde{x}) + (1-t)f(\tilde{y}). \quad (f_i \leq f, \text{ and } t, 1-t \geq 0) \end{aligned}$$

Both f_1 and f_2 at $t\tilde{x} + (1-t)\tilde{y}$ are at most $tf(\tilde{x}) + (1-t)f(\tilde{y})$, so their maximum $f(t\tilde{x} + (1-t)\tilde{y})$ is too. $\square$

Example 0.16 (A maximum is convex; a minimum need not be).

- $f_1(x) = x$ is convex, by the computation for $2x + 1$ in [Example 0.15](#) with 2 and 1 replaced by 1 and 0, and $f_2(x) = x^2$ is convex by the same example, so $\max\{x, x^2\}$ is convex by [Theorem 0.7](#).
- The minimum of two convex functions need not be convex: $g(x) = \min\{x^2, (x-2)^2\}$ has $g(0) = g(2) = 0$, but at the midpoint $g(1) = \min\{1, 1\} = 1 > \frac{1}{2}g(0) + \frac{1}{2}g(2) = 0$.

Definition 0.6 (Epigraph). The **epigraph** of $f : \mathbb{R}^p \rightarrow \mathbb{R}$ is the set of points on or over its [graph](#),

$$\text{epi } f \stackrel{\text{def}}{=} \{(\tilde{x}, s) : \tilde{x} \in \mathbb{R}^p, s \in \mathbb{R}, f(\tilde{x}) \leq s\}.$$

A pair $(\tilde{x}, s)$ is treated as the $(p+1)$ -vector with entries $x_1, \dots, x_p, s$, so $\text{epi } f \subseteq \mathbb{R}^{p+1}$, and sums and multiples of pairs are taken entry by entry.

Example 0.17 (The epigraph of x^2). For $f(x) = x^2$, $\text{epi } f = \{(x, s) : x^2 \leq s\}$, the points (x, s) on the parabola $s = x^2$ or on the side of it where s is larger. $(1, 2)$ is in it, since $1^2 \leq 2$; $(2, 1)$ is not, since $2^2 > 1$.

Theorem 0.8 (Convexity through the epigraph). *$f : \mathbb{R}^p \rightarrow \mathbb{R}$ is [convex](#) if and only if its epigraph ([Definition 0.6](#)) is convex ([Definition 0.3](#)).*

i Proof

Proof. If. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$ and $t \in [0, 1]$. The points $(\tilde{x}, f(\tilde{x}))$ and $(\tilde{y}, f(\tilde{y}))$ are in $\text{epi } f$, since $f(\tilde{x}) \leq f(\tilde{x})$ and $f(\tilde{y}) \leq f(\tilde{y})$. Since $\text{epi } f$ is convex, it contains

$$t(\tilde{x}, f(\tilde{x})) + (1-t)(\tilde{y}, f(\tilde{y})) = (t\tilde{x} + (1-t)\tilde{y}, tf(\tilde{x}) + (1-t)f(\tilde{y})),$$

and by Definition 0.6 that membership means $f(t\tilde{x} + (1-t)\tilde{y}) \leq tf(\tilde{x}) + (1-t)f(\tilde{y})$.

Only if. Let $(\tilde{x}, s)$ and $(\tilde{y}, u)$ be in $\text{epi } f$, so $f(\tilde{x}) \leq s$ and $f(\tilde{y}) \leq u$, and let $t \in [0, 1]$. Then

$$\begin{aligned} f(t\tilde{x} + (1-t)\tilde{y}) &\leq tf(\tilde{x}) + (1-t)f(\tilde{y}) \quad (f \text{ is convex}) \\ &\leq ts + (1-t)u, \quad (f(\tilde{x}) \leq s, f(\tilde{y}) \leq u, \text{ and } t, 1-t \geq 0) \end{aligned}$$

so $t(\tilde{x}, s) + (1-t)(\tilde{y}, u) = (t\tilde{x} + (1-t)\tilde{y}, ts + (1-t)u)$ is in $\text{epi } f$. $\square$

Example 0.18 (Two epigraphs).

- The epigraph of $f(x) = |x|$ is $\{(x, s) : |x| \leq s\}$, a wedge. It is the intersection of the half-planes $\{(x, s) : s - x \geq 0\}$ and $\{(x, s) : s + x \geq 0\}$. Each half-plane is convex: if $s_1 - x_1 \geq 0$ and $s_2 - x_2 \geq 0$, then $(ts_1 + (1-t)s_2) - (tx_1 + (1-t)x_2) = t(s_1 - x_1) + (1-t)(s_2 - x_2) \geq 0$, and the same with $+$ for the other. So the wedge is convex (Theorem 0.6), and $|x|$ is convex by Theorem 0.8.
- The epigraph of $g(x) = -x^2$ is not convex: $(-1, -1)$ and $(1, -1)$ are in it, since $g(\pm 1) = -1$, but their midpoint $(0, -1)$ is not, since $g(0) = 0 > -1$. So $-x^2$ is not convex.

Definition 0.7 (Tangent plane). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have a gradient at $\tilde{y}$ (gradient). The **tangent plane** to the **graph** of f at $\tilde{y}$ is the graph of the function

$$\tilde{x} \mapsto f(\tilde{y}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top (\tilde{x} - \tilde{y}),$$

which takes the value $f(\tilde{y})$ at $\tilde{x} = \tilde{y}$ and has the same gradient as f there. For $p = 1$ it is the **tangent line** $s = f(y) + f'(y)(x - y)$, the line through $(y, f(y))$ with slope $f'(y)$.

Example 0.19 (A tangent line and a tangent plane).

- For $f(x) = x^2$ at $y = 1$, $f(1) = 1$ and $f'(1) = 2$, so the tangent line is

$$\begin{aligned}
s &= 1 + 2(x - 1) && \text{(substitute } f(1) \text{ and } f'(1)) \\
&= 1 + 2x - 2 && \text{(distribute)} \\
&= 2x - 1. && (1 - 2 = -1)
\end{aligned}$$

- For $f(\tilde{x}) = x_1^2 + x_2^2$ at $\tilde{y} = (1, 1)^\top$, $f(\tilde{y}) = 2$ and the gradient is $(2y_1, 2y_2)^\top = (2, 2)^\top$, so the tangent plane is

$$\begin{aligned}
s &= 2 + 2(x_1 - 1) + 2(x_2 - 1) && \text{(substitute } f(\tilde{y}) \text{ and the gradient)} \\
&= 2 + 2x_1 - 2 + 2x_2 - 2 && \text{(distribute)} \\
&= 2x_1 + 2x_2 - 2. && (2 - 2 - 2 = -2)
\end{aligned}$$

Theorem 0.9 (First-order characterization of convexity). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$ (continuity). Then f is convex if and only if*

$$f(\tilde{x}) \geq f(\tilde{y}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top (\tilde{x} - \tilde{y}) \quad \text{for all } \tilde{x}, \tilde{y} \in \mathbb{R}^p : \quad (1)$$

every tangent plane (Definition 0.7) lies on or under the graph. If the inequality is strict whenever $\tilde{x} \neq \tilde{y}$, then f is strictly convex (Definition 0.5).

i Proof

Proof. Only if. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$, let $\tilde{d} \stackrel{\text{def}}{=} \tilde{x} - \tilde{y}$, and let $g(t) \stackrel{\text{def}}{=} f(\tilde{y} + t\tilde{d})$. The inner function $t \mapsto \tilde{y} + t\tilde{d}$ has derivative $\tilde{d}^\top$ (vector-valued derivative), so the vector chain rule (vector chain rule) gives

$$\begin{aligned}
g'(0) &= \tilde{d}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{y}) && \text{(vector chain rule)} \\
&= \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top \tilde{d}. && \text{(a } 1 \times 1 \text{ matrix equals its transpose)}
\end{aligned}$$

For $t \in (0, 1]$, $\tilde{y} + t\tilde{d} = t\tilde{x} + (1 - t)\tilde{y}$, so

$$\begin{aligned}
g(t) - g(0) &= f(t\tilde{x} + (1 - t)\tilde{y}) - f(\tilde{y}) && \text{(definition of } g) \\
&\leq tf(\tilde{x}) + (1 - t)f(\tilde{y}) - f(\tilde{y}) && (f \text{ is convex)} \\
&= t(f(\tilde{x}) - f(\tilde{y})), && \text{(collect terms)}
\end{aligned}$$

and dividing by $t > 0$, $\frac{g(t) - g(0)}{t} \leq f(\tilde{x}) - f(\tilde{y})$. As $t \rightarrow 0$ from the right, the left side tends to $g'(0)$ (g is differentiable at 0, so the one-sided limit equals the two-sided derivative), and a limit of numbers that are all at most $f(\tilde{x}) - f(\tilde{y})$ is at most $f(\tilde{x}) - f(\tilde{y})$.

So $(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}))^\top (\tilde{x} - \tilde{y}) \leq f(\tilde{x}) - f(\tilde{y})$, which rearranges to Equation 1.

If. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$, $t \in [0, 1]$, $\tilde{z} \stackrel{\text{def}}{=} t\tilde{x} + (1-t)\tilde{y}$, and $\tilde{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} f(\tilde{z})$. Applying Equation 1 at the point $\tilde{z}$, once toward $\tilde{x}$ and once toward $\tilde{y}$,

$$f(\tilde{x}) \geq f(\tilde{z}) + \tilde{g}^\top (\tilde{x} - \tilde{z}), \quad f(\tilde{y}) \geq f(\tilde{z}) + \tilde{g}^\top (\tilde{y} - \tilde{z}).$$

Multiplying the first by $t \geq 0$ and the second by $1 - t \geq 0$ and adding,

$$\begin{aligned} tf(\tilde{x}) + (1-t)f(\tilde{y}) &\geq t(f(\tilde{z}) + \tilde{g}^\top (\tilde{x} - \tilde{z})) + (1-t)(f(\tilde{z}) + \tilde{g}^\top (\tilde{y} - \tilde{z})) && \text{(multiply by } t, 1-t \geq 0 \text{ and add)} \\ &= (t + (1-t))f(\tilde{z}) + t\tilde{g}^\top (\tilde{x} - \tilde{z}) + (1-t)\tilde{g}^\top (\tilde{y} - \tilde{z}) && \text{(distribute } t \text{ and } 1-t) \\ &= f(\tilde{z}) + t\tilde{g}^\top (\tilde{x} - \tilde{z}) + (1-t)\tilde{g}^\top (\tilde{y} - \tilde{z}) && (t + (1-t) = 1) \\ &= f(\tilde{z}) + \tilde{g}^\top (t(\tilde{x} - \tilde{z}) + (1-t)(\tilde{y} - \tilde{z})) && \text{(factor out } \tilde{g}^\top; \text{ matrix products distributive)} \\ &= f(\tilde{z}) + \tilde{g}^\top (t\tilde{x} + (1-t)\tilde{y} - \tilde{z}) && \text{(collect terms)} \\ &= f(\tilde{z}). && \text{(definition of } \tilde{z}) \end{aligned}$$

That inequality is the defining inequality of a **convex function**. If Equation 1 is strict for distinct points, and $\tilde{x} \neq \tilde{y}$ and $t \in (0, 1)$, then:

- $\tilde{z}$ differs from both $\tilde{x}$ and $\tilde{y}$, since $\tilde{x} - \tilde{z} = (1-t)(\tilde{x} - \tilde{y}) \neq \tilde{0}$ and $\tilde{y} - \tilde{z} = t(\tilde{y} - \tilde{x}) \neq \tilde{0}$;
- so both inequalities at $\tilde{z}$ are strict;
- both multipliers t and $1 - t$ are positive, so the first step of the display is strict, and f is strictly convex.

□

Example 0.20 (Tangent lines under a parabola). For $f(x) = x^2$, the right side of Equation 1 is $y^2 + 2y(x - y) = 2xy - y^2$, and

$$f(x) - (2xy - y^2) = x^2 - 2xy + y^2 = (x - y)^2 \geq 0,$$

with equality only at $x = y$, confirming that x^2 is strictly convex. For $g(x) = -x^2$ the inequality fails: at $y = 0$ the tangent line is $s = 0$, and $g(1) = -1 < 0$.

Theorem 0.10 (Second-derivative test for convexity). *Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives that are continuous on $\mathbb{R}^p$ (**continuity**).*

1. If $\mathbf{H}_f(\tilde{x})$ is **positive semidefinite** for every $\tilde{x}$, then f is **convex**.
2. If $\mathbf{H}_f(\tilde{x})$ is **positive definite** for every $\tilde{x}$, then f is **strictly convex** (Definition 0.5).

i Proof

Proof. Let $\tilde{x}, \tilde{y} \in \mathbb{R}^p$. By Theorem 0.4 with $\tilde{z} = \tilde{y}$ and $\tilde{h} = \tilde{x} - \tilde{y}$, there is a $\tau \in (0, 1)$ such that, with $\tilde{w} \stackrel{\text{def}}{=} \tilde{y} + \tau(\tilde{x} - \tilde{y})$,

$$\begin{aligned} f(\tilde{x}) &= f(\tilde{y}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top (\tilde{x} - \tilde{y}) + \frac{1}{2} (\tilde{x} - \tilde{y})^\top \mathbf{H}_f(\tilde{w}) (\tilde{x} - \tilde{y}) \quad (\text{Theorem 0.4}) \\ &\geq f(\tilde{y}) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{y}) \right)^\top (\tilde{x} - \tilde{y}). \end{aligned} \quad (\mathbf{H}_f(\tilde{w}) \text{ is positive semidefinite})$$

This inequality is Equation 1, so f is convex by Theorem 0.9. Under the hypothesis of part 2 and with $\tilde{x} \neq \tilde{y}$, the quadratic-form term is positive, so the inequality is strict, and f is strictly convex by the last sentence of Theorem 0.9. $\square$

Example 0.21 (Using the Hessian to show convexity).

- For a symmetric positive semidefinite $\mathbf{A}$, $f(\tilde{x}) = \tilde{x}^\top \mathbf{A} \tilde{x}$ has Hessian $2\mathbf{A}$ (Hessian of a quadratic form), which is positive semidefinite because $\tilde{h}^\top (2\mathbf{A}) \tilde{h} = 2 \tilde{h}^\top \mathbf{A} \tilde{h} \geq 0$; so f is convex.
- $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ has Hessian $2\mathbf{I}_2$ (Example 0.9), which is positive definite, so f is strictly convex.
- $f(\tilde{x}) = e^{2x_1+x_2} - x_1$ has Hessian $e^u \begin{bmatrix} 4 & 2 \\ 2 & 1 \end{bmatrix}$ with $u = 2x_1 + x_2$ (Example), and

$$\tilde{h}^\top \mathbf{H}_f(\tilde{x}) \tilde{h} = e^u (4h_1^2 + 4h_1h_2 + h_2^2) = e^u (2h_1 + h_2)^2 \geq 0,$$

so f is convex. The Hessian is not positive definite ($\tilde{h} = (1, -2)^\top$ gives 0), so part 2 does not apply, and in fact f is not strictly convex: along $\tilde{x} = t(1, -2)^\top$, $u = 2t - 2t = 0$ and $f = 1 - t$ is linear in t , so the convexity inequality holds with equality there.

Example 0.22 (Strictly convex with a singular Hessian). Part 2 is sufficient but not necessary. $f(x) = x^4$ has $f''(0) = 12 \cdot 0^2 = 0$, so $\mathbf{H}_f(0) = [0]$ is singular and not positive definite. Yet f is strictly convex, by the strict form of Equation 1: for $x \neq y$,

$$\begin{aligned}
x^4 - y^4 - 4y^3(x - y) &= (x - y)(x^3 + x^2y + xy^2 + y^3) - 4y^3(x - y) && \text{(factor } x^4 - y^4\text{)} \\
&= (x - y)(x^3 + x^2y + xy^2 - 3y^3) && \text{(collect the } y^3 \text{ terms)} \\
&= (x - y)^2(x^2 + 2xy + 3y^2) && ((x - y)(x^2 + 2xy + 3y^2) = x^3 + x^2y + x \\
&= (x - y)^2((x + y)^2 + 2y^2), && \text{(complete the square)}
\end{aligned}$$

which is positive: $(x - y)^2 > 0$, and $(x + y)^2 + 2y^2 = 0$ would need $y = 0$ and then $x = 0 = y$.

Corollary 0.1 (Stationary points of a convex function are global minimizers). *If f is as in Theorem 0.9 and convex, then every stationary point $\tilde{x}^*$ of f (Definition 0.1) is a global minimizer.*

i Proof

Proof. For every $\tilde{x}$,

$$\begin{aligned}
f(\tilde{x}) &\geq f(\tilde{x}^*) + \left(\frac{\partial}{\partial \tilde{x}} f(\tilde{x}^*) \right)^\top (\tilde{x} - \tilde{x}^*) && \text{(Equation 1 with } \tilde{y} = \tilde{x}^*) \\
&= f(\tilde{x}^*) + \tilde{0}^\top (\tilde{x} - \tilde{x}^*) && (\tilde{x}^* \text{ is stationary)} \\
&= f(\tilde{x}^*). && (\tilde{0}^\top \tilde{v} = 0)
\end{aligned}$$

□

Example 0.23 (A convex function, and a non-convex one).

- $f(\tilde{x}) = x_1^2 + x_2^2 - 2x_1$ is convex by Theorem 0.10, since its Hessian is $2\mathbf{I}_2$ (Example 0.9), so its stationary point $(1, 0)^\top$ (Example 0.2) is a global minimizer, as completing the square showed in Example 0.3.
- Without convexity, a stationary point need not be a minimizer at all: $x_1^2 - x_2^2$ has the stationary point $\tilde{0}$, which is not a minimizer (Example 0.3).
- The corollary says nothing about whether a minimizer exists: $e^{2x_1+x_2} - x_1$ is convex (Example 0.21) but has no stationary point, and so no local minimizer (Example 0.3).

Minimizing a quadratic under linear constraints

Source

This section is adapted from the Lagrange-multiplier part of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source solves the stationarity equations of the Lagrangian and reports the solution; here the solution is also proved to be the unique minimizer, directly from the positive definiteness of the matrices involved. The source assumes only that $\mathbf{S}$ is positive definite; the added hypothesis that the columns of $\mathbf{A}$ are linearly independent makes $\mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A}$ invertible. The source's example with general coefficients a_1, a_2, b is specialized here to 3, 4, 5. These parts of the source are not part of this section:

- the Lagrange multiplier method for constraints that are not linear, or objectives that are not quadratic, including its multinomial maximum likelihood example
- the eigenvalues of the block matrix of the stationarity equations, and the convexity of the Lagrangian in $\tilde{x}$ and its concavity in the multipliers
- its homework examples

Definition 0.8 (Feasible set and constraints). In a problem that looks for a best point among some allowed points of $\mathbb{R}^p$, the set $F \subseteq \mathbb{R}^p$ of allowed points is the **feasible set**, and its points are **feasible**. When F is the set of points that satisfy some equations or inequalities, those equations or inequalities are the **constraints**.

Example 0.24 (A line as a feasible set). With the single constraint $3x_1 + 4x_2 = 5$, the feasible set is the line $\{\tilde{x} \in \mathbb{R}^2 : 3x_1 + 4x_2 = 5\}$. $(1, 0.5)^\top$ is feasible, since $3 \cdot 1 + 4 \cdot 0.5 = 3 + 2 = 5$; $\tilde{0}$ is not feasible, since $3 \cdot 0 + 4 \cdot 0 = 0 \neq 5$.

Definition 0.9 (Objective function (objective) and minimization problem). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ and let $F \subseteq \mathbb{R}^p$ be a feasible set (Definition 0.8). In the problem of minimizing f over F , f is the **objective function** (or **objective**). A **minimizer of f over F** is a feasible $\tilde{x}^*$ with $f(\tilde{x}^*) \leq f(\tilde{x})$ for every feasible $\tilde{x}$: a [global minimizer](#) of f with its domain restricted to F . When F is given by constraints, the problem is to minimize f **subject to** them.

Example 0.25 (Minimizing over a line). In the problem of minimizing $x_1^2 + x_2^2$ subject to $3x_1 + 4x_2 = 5$, the objective function is $f(\tilde{x}) = x_1^2 + x_2^2$, and the feasible set is the line of Example 0.24. The feasible point $(1, 0.5)^\top$ has $f(1, 0.5) = 1 + 0.25 = 1.25$. $\tilde{0}$ minimizes f over all of $\mathbb{R}^2$, with $f(\tilde{0}) = 0$, but it is not a minimizer over the feasible set, because it is not feasible.

Here the objective (Definition 0.9) is a quadratic $\frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x}$ with $\mathbf{S}$ positive definite, and the constraints (Definition 0.8) are m linear equations $\mathbf{A}^\top \tilde{x} = \tilde{b}$, where $\mathbf{A}$ is a $p \times m$ matrix and $\tilde{b} \in \mathbb{R}^m$. The only stationary point of the objective alone (Definition 0.1) is $\tilde{0}_{p \times 1}$ (its gradient is $\mathbf{S} \tilde{x}$, and $\mathbf{S}$ is invertible), which may not satisfy the constraints. A Lagrange multiplier adds one unknown per constraint, and the minimizer over the feasible set (Definition 0.8) is then found among the stationary points of a new function of the extended set of unknowns.

Definition 0.10 (Lagrangian). For the problem of minimizing $f(\tilde{x})$ over $\tilde{x} \in \mathbb{R}^p$ subject to $\mathbf{A}^\top \tilde{x} = \tilde{b}$, with $\mathbf{A}$ a $p \times m$ matrix and $\tilde{b} \in \mathbb{R}^m$, the **Lagrangian** is the function of $\tilde{x} \in \mathbb{R}^p$ and $\tilde{\lambda} \in \mathbb{R}^m$

$$L(\tilde{x}, \tilde{\lambda}) \stackrel{\text{def}}{=} f(\tilde{x}) + \tilde{\lambda}^\top (\mathbf{A}^\top \tilde{x} - \tilde{b}),$$

and the entries of $\tilde{\lambda}$ are the **Lagrange multipliers**, one per constraint.

Example 0.26 (The gradients of the Lagrangian of a quadratic). Let $f(\tilde{x}) = \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x}$ with $\mathbf{S}$ symmetric and constant. By the [transpose of a product](#), $\tilde{\lambda}^\top \mathbf{A}^\top = (\mathbf{A} \tilde{\lambda})^\top$, so

$$\begin{aligned} L(\tilde{x}, \tilde{\lambda}) &= \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x} + \tilde{\lambda}^\top \mathbf{A}^\top \tilde{x} - \tilde{\lambda}^\top \tilde{b} \quad (\text{distribute}) \\ &= \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x} + (\mathbf{A} \tilde{\lambda})^\top \tilde{x} - \tilde{\lambda}^\top \tilde{b}, \quad (\text{transpose of a product}) \end{aligned}$$

and, since $\tilde{\lambda}^\top (\mathbf{A}^\top \tilde{x} - \tilde{b})$ is 1×1 and so equals its transpose,

$$L(\tilde{x}, \tilde{\lambda}) = \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x} + (\mathbf{A}^\top \tilde{x} - \tilde{b})^\top \tilde{\lambda}.$$

The partial derivatives of a sum are the sums of the partial derivatives, entry by entry. Using the first of these forms, [derivative of a quadratic form](#) for the first term, [derivative of a linear combination](#) for the second ($\mathbf{A} \tilde{\lambda}$ is constant in $\tilde{x}$), and [constant with respect to a vector](#) for the third,

$$\begin{aligned} \frac{\partial}{\partial \tilde{x}} L(\tilde{x}, \tilde{\lambda}) &= \frac{1}{2} \frac{\partial}{\partial \tilde{x}} (\tilde{x}^\top \mathbf{S} \tilde{x}) + \frac{\partial}{\partial \tilde{x}} ((\mathbf{A} \tilde{\lambda})^\top \tilde{x}) - \frac{\partial}{\partial \tilde{x}} (\tilde{\lambda}^\top \tilde{b}) \quad (\text{sum and constant multiple}) \\ &= \frac{1}{2} (2\mathbf{S} \tilde{x}) + \mathbf{A} \tilde{\lambda} - \tilde{0}_{p \times 1} \quad (\text{the three derivatives}) \\ &= \mathbf{S} \tilde{x} + \mathbf{A} \tilde{\lambda}. \quad (\text{simplify}) \end{aligned}$$

Using the second form, in which $f(\tilde{x})$ and $\mathbf{A}^\top \tilde{x} - \tilde{b}$ are constant in $\tilde{\lambda}$, and [derivative of a linear combination](#) again,

$$\begin{aligned}
\frac{\partial}{\partial \tilde{\lambda}} L(\tilde{x}, \tilde{\lambda}) &= \frac{\partial}{\partial \tilde{\lambda}} f(\tilde{x}) + \frac{\partial}{\partial \tilde{\lambda}} \left((\mathbf{A}^\top \tilde{x} - \tilde{b})^\top \tilde{\lambda} \right) \quad (\text{sum}) \\
&= \tilde{0}_{m \times 1} + (\mathbf{A}^\top \tilde{x} - \tilde{b}) \quad (\text{the two derivatives}) \\
&= \mathbf{A}^\top \tilde{x} - \tilde{b}. \quad (\text{simplify})
\end{aligned}$$

So $(\tilde{x}, \tilde{\lambda})$ makes the gradients equal $\tilde{0}_{p \times 1}$ and $\tilde{0}_{m \times 1}$ exactly when

$$\mathbf{S}\tilde{x} + \mathbf{A}\tilde{\lambda} = \tilde{0}_{p \times 1} \quad \text{and} \quad \mathbf{A}^\top \tilde{x} = \tilde{b}, \quad \text{that is,} \quad \underbrace{\begin{bmatrix} \mathbf{S} & \mathbf{A} \\ \mathbf{A}^\top & \tilde{0}_{m \times m} \end{bmatrix}}_{(p+m) \times (p+m)} \underbrace{\begin{bmatrix} \tilde{x} \\ \tilde{\lambda} \end{bmatrix}}_{(p+m) \times 1} = \underbrace{\begin{bmatrix} \tilde{0}_{p \times 1} \\ \tilde{b} \end{bmatrix}}_{(p+m) \times 1}, \quad (2)$$

the second equation being the constraint itself.

Theorem 0.11 (Minimizing a positive definite quadratic under linear constraints). *Let $\mathbf{S}$ be a $p \times p$ [positive definite](#) matrix, $\mathbf{A}$ a $p \times m$ matrix with [linearly independent](#) columns, and $\tilde{b} \in \mathbb{R}^m$. Then the $m \times m$ matrix $\mathbf{M} \stackrel{\text{def}}{=} \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A}$ is invertible, and $f(\tilde{x}) = \frac{1}{2} \tilde{x}^\top \mathbf{S} \tilde{x}$ has exactly one minimizer over the $\tilde{x} \in \mathbb{R}^p$ with $\mathbf{A}^\top \tilde{x} = \tilde{b}$, namely*

$$\tilde{x}^* = \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b}.$$

With $\tilde{\lambda}^* \stackrel{\text{def}}{=} -\mathbf{M}^{-1} \tilde{b}$, the pair $(\tilde{x}^*, \tilde{\lambda}^*)$ solves Equation 2, and the minimum value is $f(\tilde{x}^*) = \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \tilde{b}$.

i Proof

Proof. The steps marked “transpose of a product” use the [transpose of a product](#), and $\mathbf{S}$ is symmetric because it is [positive definite](#).

M is invertible. $\mathbf{S}^{-1}$ exists and is positive definite ([positive definite inverse](#)), so $\mathbf{M} = \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A}$ is positive definite ([operations that preserve definiteness](#), part 1), and so invertible (positive definite inverse again). $\mathbf{S}^{-1}$ and $\mathbf{M}^{-1}$ are symmetric ([inverse of a symmetric matrix](#)), because $\mathbf{S}$ and $\mathbf{M}$ are.

$\tilde{x}^*$ satisfies the constraint, and the pair solves Equation 2.

$$\begin{aligned}
\mathbf{A}^\top \tilde{x}^* &= \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(substitute } \tilde{x}^*) \\
&= \mathbf{M} \mathbf{M}^{-1} \tilde{b} && \text{(definition of } \mathbf{M}) \\
&= \tilde{b}, && (\mathbf{M} \mathbf{M}^{-1} = \mathbf{I}_m) \\
\mathbf{S} \tilde{x}^* + \mathbf{A} \tilde{\lambda}^* &= \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} - \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(substitute } \tilde{x}^* \text{ and } \tilde{\lambda}^*) \\
&= \mathbf{A} \mathbf{M}^{-1} \tilde{b} - \mathbf{A} \mathbf{M}^{-1} \tilde{b} && (\mathbf{S} \mathbf{S}^{-1} = \mathbf{I}_p) \\
&= \tilde{0}_{p \times 1}. && \text{(subtract)}
\end{aligned}$$

$\tilde{x}^*$ is the unique minimizer. Let $\tilde{x}$ satisfy $\mathbf{A}^\top \tilde{x} = \tilde{b}$, and let $\tilde{d} \stackrel{\text{def}}{=} \tilde{x} - \tilde{x}^*$, so $\mathbf{A}^\top \tilde{d} = \mathbf{A}^\top \tilde{x} - \mathbf{A}^\top \tilde{x}^* = \tilde{b} - \tilde{b} = \tilde{0}_{m \times 1}$ (matrix products distribute; both points satisfy the constraint). First, the cross term vanishes:

$$\begin{aligned}
\tilde{d}^\top \mathbf{S} \tilde{x}^* &= \tilde{d}^\top \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(substitute } \tilde{x}^*) \\
&= \tilde{d}^\top \mathbf{A} \mathbf{M}^{-1} \tilde{b} && (\mathbf{S} \mathbf{S}^{-1} = \mathbf{I}_p) \\
&= (\mathbf{A}^\top \tilde{d})^\top \mathbf{M}^{-1} \tilde{b} && \text{(transpose of a product; } (\mathbf{A}^\top)^\top = \mathbf{A}) \\
&= \tilde{0}_{m \times 1}^\top \mathbf{M}^{-1} \tilde{b} && (\mathbf{A}^\top \tilde{d} = \tilde{0}_{m \times 1}) \\
&= 0. && \text{(a product with a zero factor is zero)}
\end{aligned}$$

Then

$$\begin{aligned}
f(\tilde{x}) &= \frac{1}{2} (\tilde{x}^* + \tilde{d})^\top \mathbf{S} (\tilde{x}^* + \tilde{d}) && (\tilde{x} = \tilde{x}^* + \tilde{d}) \\
&= \frac{1}{2} (\tilde{x}^{*\top} + \tilde{d}^\top) \mathbf{S} (\tilde{x}^* + \tilde{d}) && \text{(the transpose of a sum is the sum of the transposes)} \\
&= \frac{1}{2} (\tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + \tilde{x}^{*\top} \mathbf{S} \tilde{d} + \tilde{d}^\top \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{d}) && \text{(distribute)} \\
&= \frac{1}{2} (\tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S}^\top \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{d}) && (1 \times 1: \tilde{x}^{*\top} \mathbf{S} \tilde{d} = (\tilde{x}^{*\top} \mathbf{S} \tilde{d})^\top; \text{transpose of a product}) \\
&= \frac{1}{2} (\tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + 2 \tilde{d}^\top \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{d}) && (\mathbf{S}^\top = \mathbf{S}; \text{combine}) \\
&= \frac{1}{2} \tilde{x}^{*\top} \mathbf{S} \tilde{x}^* + \tilde{d}^\top \mathbf{S} \tilde{x}^* + \frac{1}{2} \tilde{d}^\top \mathbf{S} \tilde{d} && \text{(distribute } \frac{1}{2}) \\
&= f(\tilde{x}^*) + \frac{1}{2} \tilde{d}^\top \mathbf{S} \tilde{d} && \text{(the cross term is 0)} \\
&\geq f(\tilde{x}^*), && (\mathbf{S} \text{ is positive definite})
\end{aligned}$$

with equality only when $\tilde{d} = \tilde{0}_{p \times 1}$, that is, $\tilde{x} = \tilde{x}^*$.

The minimum value.

$$\begin{aligned}
f(\tilde{x}^*) &= \frac{1}{2} (\mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b})^\top \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(substitute } \tilde{x}^*) \\
&= \frac{1}{2} \tilde{b}^\top (\mathbf{M}^{-1})^\top \mathbf{A}^\top (\mathbf{S}^{-1})^\top \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && \text{(transpose of a product, applied repeatedly)} \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{S} \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && (\mathbf{S}^{-1} \text{ and } \mathbf{M}^{-1} \text{ are symmetric}) \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \mathbf{A}^\top \mathbf{S}^{-1} \mathbf{A} \mathbf{M}^{-1} \tilde{b} && (\mathbf{S}^{-1} \mathbf{S} = \mathbf{I}_p) \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \mathbf{M} \mathbf{M}^{-1} \tilde{b} && \text{(definition of } \mathbf{M}) \\
&= \frac{1}{2} \tilde{b}^\top \mathbf{M}^{-1} \tilde{b}. && (\mathbf{M}^{-1} \mathbf{M} = \mathbf{I}_m)
\end{aligned}$$

□

Example 0.27 (The closest point on a line to the origin). Minimize $x_1^2 + x_2^2$ subject to $3x_1 + 4x_2 = 5$. In the form of Theorem 0.11, $x_1^2 + x_2^2 = \frac{1}{2} \tilde{x}^\top (2\mathbf{I}_2) \tilde{x}$, so $\mathbf{S} = 2\mathbf{I}_2$, $\mathbf{A} = (3, 4)^\top$ ($p = 2$, $m = 1$) and $\tilde{b} = [5]$. The hypotheses hold: $2\mathbf{I}_2$ is positive definite, and the single column $(3, 4)^\top$ is nonzero, so it is linearly independent. Also $\mathbf{S}^{-1} = \frac{1}{2}\mathbf{I}_2$. Then, from the formulas of Theorem 0.11,

$$\begin{aligned}
\mathbf{M} &= \mathbf{A}^\top \frac{1}{2} \mathbf{I}_2 \mathbf{A} = \frac{1}{2} (3^2 + 4^2) = \frac{25}{2}, \\
\tilde{x}^* &= \frac{1}{2} \mathbf{I}_2 \begin{bmatrix} 3 \\ 4 \end{bmatrix} \left(\frac{2}{25} \right) (5) = \frac{5}{25} \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 0.6 \\ 0.8 \end{bmatrix}, \\
\tilde{\lambda}^* &= -\left(\frac{2}{25} \right) (5) = -0.4, \\
f(\tilde{x}^*) &= \frac{1}{2} (5) \left(\frac{2}{25} \right) (5) = 1.
\end{aligned}$$

Checks:

- the constraint: $3(0.6) + 4(0.8) = 1.8 + 3.2 = 5$;
- the first block of Equation 2: $\mathbf{S}\tilde{x}^* + \mathbf{A}\tilde{\lambda}^* = (1.2, 1.6)^\top + (3, 4)^\top (-0.4) = (1.2, 1.6)^\top - (1.2, 1.6)^\top = \tilde{0}_{2 \times 1}$;
- the value: $0.6^2 + 0.8^2 = 0.36 + 0.64 = 1$.

Geometrically, $\tilde{x}^*$ is the point of the line nearest the origin, at distance 1.

Example 0.28 (The hypotheses are needed).

- If $\mathbf{S}$ is not positive definite there may be no minimizer: with $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ and the constraint $x_1 = 1$ ($\mathbf{A} = (1, 0)^\top$, $\tilde{b} = [1]$), $f(1, x_2) = \frac{1}{2}(1 - x_2^2)$ decreases without bound as x_2 grows.

- If the columns of $\mathbf{A}$ are dependent, $\mathbf{M}$ is not invertible: with $\mathbf{S} = \mathbf{I}_2$ and $\mathbf{A} = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$, $\mathbf{M} = \mathbf{A}^\top \mathbf{A} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$, whose columns are equal. So $\mathbf{M}(1, -1)^\top = \tilde{\mathbf{0}}_{2 \times 1}$, and $\mathbf{M}$ is not invertible (a square matrix is invertible exactly when its null space is zero). The two constraints both read $x_1 + x_2 = b_i$, so they have no solution at all when $b_1 \neq b_2$. When $b_1 = b_2$ a unique minimizer still exists, $x_1 = x_2 = b_1/2$, but the formula of Theorem 0.11 cannot produce it.

Definition 0.11 (Sensitivity). Suppose a minimization problem (Definition 0.9) depends on a vector $\tilde{\mathbf{b}} \in \mathbb{R}^m$, and let $f^*(\tilde{\mathbf{b}})$ be its minimum value, as a function of $\tilde{\mathbf{b}}$. When the gradient $\frac{\partial}{\partial \tilde{\mathbf{b}}} f^*(\tilde{\mathbf{b}})$ exists (gradient), it is the **sensitivity** of the minimum value to $\tilde{\mathbf{b}}$: by **partial derivative**, its entry i is the limit, as a change in b_i alone shrinks to 0, of the resulting change in the minimum value divided by the change in b_i .

Example 0.29 (The sensitivity of a one-point problem). Minimize x^2 subject to $x = b$, for a number b . The feasible set (Definition 0.8) is $\{b\}$, so the minimum value is $f^*(b) = b^2$, and its sensitivity to b is $\frac{d}{db} f^*(b) = 2b$. At $b = 3$ the sensitivity is 6: moving b from 3 to 3.1 raises the minimum value from 9 to 9.61, an increase of 0.61, close to $6 \times 0.1 = 0.6$.

Corollary 0.2 (The multiplier is the sensitivity of the minimum value). In Theorem 0.11, write the minimum value as a function of $\tilde{\mathbf{b}}$, $f^*(\tilde{\mathbf{b}}) = \frac{1}{2} \tilde{\mathbf{b}}^\top \mathbf{M}^{-1} \tilde{\mathbf{b}}$. Its sensitivity to $\tilde{\mathbf{b}}$ (Definition 0.11) is

$$\frac{\partial}{\partial \tilde{\mathbf{b}}} f^*(\tilde{\mathbf{b}}) = \mathbf{M}^{-1} \tilde{\mathbf{b}} = -\tilde{\lambda}^*.$$

Proof

Proof. $\mathbf{M}^{-1}$ is symmetric and does not depend on $\tilde{\mathbf{b}}$, so **derivative of a quadratic form** applies with $\mathbf{M}^{-1}$ in place of $\mathbf{S}$:

$$\begin{aligned} \frac{\partial}{\partial \tilde{\mathbf{b}}} f^*(\tilde{\mathbf{b}}) &= \frac{1}{2} \frac{\partial}{\partial \tilde{\mathbf{b}}} (\tilde{\mathbf{b}}^\top \mathbf{M}^{-1} \tilde{\mathbf{b}}) && \text{(constant multiple)} \\ &= \frac{1}{2} (2\mathbf{M}^{-1} \tilde{\mathbf{b}}) && \text{(derivative of a quadratic form)} \\ &= \mathbf{M}^{-1} \tilde{\mathbf{b}} && \text{(simplify)} \\ &= -\tilde{\lambda}^*. && \text{(definition of } \tilde{\lambda}^*) \end{aligned}$$

□

Example 0.30 (Moving the line). In Example 0.27 with a general right side b in place of 5, $\mathbf{M}^{-1} = \frac{2}{25}$ and $f^*(b) = \frac{1}{2}(\frac{2}{25})b^2 = \frac{b^2}{25}$, so $\frac{d}{db}f^*(b) = \frac{2b}{25}$. At $b = 5$ this derivative is $\frac{10}{25} = 0.4 = -\lambda^*$: moving the line $3x_1 + 4x_2 = b$ outward by a small step ε in b raises the minimum value by about 0.4ε .

Newton's method and gradient descent

Source

This section is adapted from the Newton-Raphson and gradient descent parts of Zhou (2024), used under the MIT License (see the license text in [Linear Algebra](#)). The source motivates the Newton step by minimizing a quadratic approximation; here that step is proved to be the exact minimizer of the quadratic model. In the source's zigzag example the starting value of the objective is $\frac{1}{2}(b^2 + b)$, not $\frac{1}{2}(1 + b^2)$ as the source writes; the ratio of successive values is unaffected. In the same example the source's formula for $\tilde{x}^{(1)}$ has $\tilde{x}^{(1)}$ on its right side where $\tilde{x}^{(0)}$ is meant. These parts of the source are not part of this section:

- the quadratic convergence theorem for Newton's method in general, and its remedies for instability (a positive definite substitute for the Hessian, and shortening the Newton step by a search along its direction)
- its remark that Hessians are expensive to compute
- the convergence-rate theorem for strongly convex functions
- gradient descent with momentum, and its optimal parameters
- Nesterov acceleration, and stochastic gradient descent
- its code and animations

Definition 0.12 (Iterates and convergence of a sequence of vectors). A [sequence](#) $\tilde{x}^{(0)}, \tilde{x}^{(1)}, \dots$ in $\mathbb{R}^p$ **converges** to $\tilde{x}^* \in \mathbb{R}^p$ if the sequence of distances $\|\tilde{x}^{(t)} - \tilde{x}^*\|$ [converges](#) to 0. For $p = 1$, $\|x^{(t)} - x^*\| = |x^{(t)} - x^*|$, and this is the convergence of a sequence of numbers to x^* .

An **iterative method** starts from a point $\tilde{x}^{(0)}$ and computes $\tilde{x}^{(t+1)}$ from $\tilde{x}^{(t)}$ by a fixed rule, for $t = 0, 1, 2, \dots$; the points $\tilde{x}^{(t)}$ are its **iterates**. For a point $\tilde{x}^*$ that the method is meant to reach, $\|\tilde{x}^{(t)} - \tilde{x}^*\|$ is the **error** of the iterate $\tilde{x}^{(t)}$.

Example 0.31 (Halving a vector). The iterative method $\tilde{x}^{(t+1)} = \frac{1}{2}\tilde{x}^{(t)}$ from $\tilde{x}^{(0)} = (4, 2)^\top$ has iterates $(4, 2)^\top, (2, 1)^\top, (1, 0.5)^\top, \dots$, that is, $\tilde{x}^{(t)} = 2^{-t}(4, 2)^\top$. With $\tilde{x}^* = \tilde{0}$, the error of $\tilde{x}^{(t)}$ is

$$\begin{aligned}\|2^{-t}(4, 2)^\top - \tilde{0}\| &= 2^{-t}\|(4, 2)^\top\| \quad (\text{norm of a multiple}) \\ &= 2^{-t}\sqrt{20}, \quad (4^2 + 2^2 = 20)\end{aligned}$$

which converges to 0, so the iterates converge to $\tilde{0}$.

Definition 0.13 (Newton's method for minimization). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first and second partial derivatives, and let $\tilde{x}^{(0)}$ be a starting point. **Newton's method** computes, for $t = 0, 1, 2, \dots$, as long as $\mathbf{H}_f(\tilde{x}^{(t)})$ is invertible,

$$\tilde{x}^{(t+1)} \stackrel{\text{def}}{=} \tilde{x}^{(t)} - [\mathbf{H}_f(\tilde{x}^{(t)})]^{-1} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)}).$$

The vector $-[\mathbf{H}_f(\tilde{x}^{(t)})]^{-1} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)})$ is the **Newton step** at $\tilde{x}^{(t)}$.

Example 0.32 (Newton's method for a cubic). Let $f(x) = \frac{1}{3}x^3 - 4x$, so $f'(x) = x^2 - 4$ and $f''(x) = 2x$. For $x^{(t)} \neq 0$, one Newton step gives

$$\begin{aligned}x^{(t+1)} &= x^{(t)} - \frac{(x^{(t)})^2 - 4}{2x^{(t)}} && (\text{Newton's method with } p = 1) \\ &= \frac{2(x^{(t)})^2 - (x^{(t)})^2 + 4}{2x^{(t)}} && (\text{common denominator}) \\ &= \frac{1}{2} \left(x^{(t)} + \frac{4}{x^{(t)}} \right). && (\text{simplify})\end{aligned}$$

From $x^{(0)} = 2.5$:

- $x^{(1)} = \frac{1}{2}(2.5 + 1.6) = 2.05$;
- $x^{(2)} = \frac{1}{2}(2.05 + 1.951219 \dots) = 2.000610$ (to six decimals);
- $x^{(3)} = 2.0000001$ (to seven decimals).

The iterates (Definition 0.12) approach 2, the local minimizer of f (a stationary point, since $f'(2) = 0$, with $f''(2) = 4 > 0$; Theorem 0.5).

Definition 0.14 (Quadratic model (second-order Taylor polynomial)). Let f be as in Definition 0.13 and let $\tilde{x}$ be a point, with $\tilde{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} f(\tilde{x})$ and $\mathbf{H} \stackrel{\text{def}}{=} \mathbf{H}_f(\tilde{x})$. The **quadratic model** of f at $\tilde{x}$ is the function of a step $\tilde{\delta} \in \mathbb{R}^p$

$$q(\tilde{\delta}) \stackrel{\text{def}}{=} f(\tilde{x}) + \tilde{g}^\top \tilde{\delta} + \frac{1}{2} \tilde{\delta}^\top \mathbf{H} \tilde{\delta},$$

also called the **second-order Taylor polynomial** of f at $\tilde{x}$: when the second partial derivatives of f are continuous, Theorem 0.4 says that $f(\tilde{x} + \tilde{\delta})$ equals the same expression

with the Hessian evaluated at a point between $\tilde{x}$ and $\tilde{x} + \tilde{\delta}$.

Example 0.33 (The model at $x = 2.5$). For $f(x) = \frac{1}{3}x^3 - 4x$ at $x = 2.5$ (Example 0.32), $g = 2.5^2 - 4 = 2.25$ and $H = [2 \cdot 2.5] = [5]$, so $q(\delta) = f(2.5) + 2.25\delta + 2.5\delta^2$. For $f(\tilde{x}) = \frac{1}{2}(x_1^2 + 3x_2^2)$, whose Hessian is $\begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ at every point, the model at any $\tilde{x}$ equals $f(\tilde{x} + \tilde{\delta})$ exactly (Example 0.5 with $\mathbf{S} = \begin{bmatrix} 1/2 & 0 \\ 0 & 3/2 \end{bmatrix}$).

Theorem 0.12 (The Newton step minimizes the quadratic model). *Let q be the quadratic model of f at $\tilde{x}$ (Definition 0.14), with $\mathbf{H} = \mathbf{H}_f(\tilde{x})$ positive definite. Its unique minimizer is the Newton step $\tilde{\delta}^* = -\mathbf{H}^{-1}\tilde{g}$.*

Proof

Proof. $\mathbf{H}$ is symmetric, by the definition of positive definite, and invertible (positive definite inverse), so $\tilde{\delta}^*$ exists and $\mathbf{H}\tilde{\delta}^* = -\tilde{g}$. For any $\tilde{d} \in \mathbb{R}^p$, first

$$\begin{aligned}\tilde{d}^\top \mathbf{H}\tilde{\delta}^* &= -\tilde{d}^\top \tilde{g} \quad (\mathbf{H}\tilde{\delta}^* = -\tilde{g}) \\ &= -\tilde{g}^\top \tilde{d}, \quad (\text{a } 1 \times 1 \text{ matrix equals its transpose})\end{aligned}$$

and $\tilde{\delta}^{*\top} \mathbf{H}\tilde{d} = \tilde{d}^\top \mathbf{H}\tilde{\delta}^*$ in the same way, using $\mathbf{H}^\top = \mathbf{H}$. Then

$$\begin{aligned}q(\tilde{\delta}^* + \tilde{d}) &= f(\tilde{x}) + \tilde{g}^\top (\tilde{\delta}^* + \tilde{d}) + \frac{1}{2} (\tilde{\delta}^* + \tilde{d})^\top \mathbf{H} (\tilde{\delta}^* + \tilde{d}) && \text{(definition of } q) \\ &= f(\tilde{x}) + \tilde{g}^\top \tilde{\delta}^* + \tilde{g}^\top \tilde{d} + \frac{1}{2} (\tilde{\delta}^{*\top} \mathbf{H}\tilde{\delta}^* + \tilde{\delta}^{*\top} \mathbf{H}\tilde{d} + \tilde{d}^\top \mathbf{H}\tilde{\delta}^* + \tilde{d}^\top \mathbf{H}\tilde{d}) && \text{(distribute)} \\ &= f(\tilde{x}) + \tilde{g}^\top \tilde{\delta}^* + \tilde{g}^\top \tilde{d} + \frac{1}{2} \tilde{\delta}^{*\top} \mathbf{H}\tilde{\delta}^* - \tilde{g}^\top \tilde{d} + \frac{1}{2} \tilde{d}^\top \mathbf{H}\tilde{d} && \text{(both cross terms equal } -\tilde{g}^\top \tilde{d}) \\ &= f(\tilde{x}) + \tilde{g}^\top \tilde{\delta}^* + \frac{1}{2} \tilde{\delta}^{*\top} \mathbf{H}\tilde{\delta}^* + \frac{1}{2} \tilde{d}^\top \mathbf{H}\tilde{d} && \text{(cancel } \tilde{g}^\top \tilde{d}) \\ &= q(\tilde{\delta}^*) + \frac{1}{2} \tilde{d}^\top \mathbf{H}\tilde{d}, && \text{(definition of } q)\end{aligned}$$

which is greater than $q(\tilde{\delta}^*)$ for every $\tilde{d} \neq \tilde{0}_{p \times 1}$, since $\mathbf{H}$ is positive definite. $\square$

Example 0.34 (The Newton step at $x = 2.5$). For the model $q(\delta) = f(2.5) + 2.25\delta + 2.5\delta^2$ of Example 0.33, $H = [5]$ is positive definite, so by Theorem 0.12 its minimizer is $\delta^* = -H^{-1}g = -2.25/5 = -0.45$. Setting the derivative $2.25 + 5\delta$ to 0 gives the same δ^* , and $2.5 - 0.45 = 2.05 = x^{(1)}$ (Example 0.32).

Definition 0.15 (Quadratic convergence). A [sequence](#) $\tilde{x}^{(0)}, \tilde{x}^{(1)}, \dots$ in $\mathbb{R}^p$ that converges (Definition 0.12) to $\tilde{x}^*$ **converges quadratically** if there is a constant C with

$$\|\tilde{x}^{(t+1)} - \tilde{x}^*\| \leq C \|\tilde{x}^{(t)} - \tilde{x}^*\|^2 \quad \text{for every } t.$$

Example 0.35 (Halving the error is not quadratic convergence). $x^{(t)} = 2^{-t}$ converges to 0, but it does not converge quadratically: the bound would need $2^{-(t+1)} \leq C (2^{-t})^2 = C 2^{-2t}$, that is, $2^{t-1} \leq C$, which fails once $t > 1 + \log_2 C$. Each step only halves the error.

Example 0.36 (The error squares at each step). For the iteration $x^{(t+1)} = \frac{1}{2}(x^{(t)} + 4/x^{(t)})$ of Example 0.32,

$$\begin{aligned} x^{(t+1)} - 2 &= \frac{1}{2} \left(x^{(t)} + \frac{4}{x^{(t)}} \right) - 2 \quad (\text{the iteration}) \\ &= \frac{(x^{(t)})^2 + 4 - 4x^{(t)}}{2x^{(t)}} \quad (\text{common denominator}) \\ &= \frac{(x^{(t)} - 2)^2}{2x^{(t)}}. \quad (\text{the numerator is a perfect square}) \end{aligned}$$

So near 2 the error is roughly squared and divided by 4 at each step: the errors are 0.05, then 0.00061, then 0.000000093, and $0.05^2/(2 \cdot 2.05) \approx 0.00061$. The iterates converge quadratically (Definition 0.15): $C = \frac{1}{2}$ works for every $x^{(t)} \geq 1$, since then $\frac{1}{2x^{(t)}} \leq \frac{1}{2}$, and every iterate from 2.5 is at least 2, since the identity makes $x^{(t+1)} - 2 \geq 0$ whenever $x^{(t)} > 0$. Zhou (2024) states such a bound for Newton's method in general; it holds from starting points close enough to a minimizer where the Hessian is positive definite, when near that minimizer the Hessian changes by at most a constant times the distance moved, which these notes do not prove.

Example 0.37 (Newton's method can find a maximum). From $x^{(0)} = -2.5$ the same iteration gives $-2.05, -2.000610, -2.0000001$, approaching -2 . But -2 is not a minimizer of f : it is a stationary point of $-f$, whose second derivative there is $-f''(-2) = 4 > 0$, so it is a strict local minimizer of $-f$ (Theorem 0.5), that is, f is strictly larger at -2 than at all nearby points. Newton's method looks only for a stationary point, and here the Hessian $f''(x) = 2x$ is negative along the way, so the quadratic model has no minimizer (with $H < 0$, $q(\delta) = f + g\delta + \frac{1}{2}H\delta^2$ decreases without bound) and Theorem 0.12 does not apply: the positive definite hypothesis there is needed.

Theorem 0.13 (Newton's method minimizes a quadratic in one step). *Let $\mathbf{S}$ be a $p \times p$ positive definite matrix, $\tilde{c} \in \mathbb{R}^p$, and $f(\tilde{x}) = \frac{1}{2}\tilde{x}^\top \mathbf{S}\tilde{x} - \tilde{c}^\top \tilde{x}$. From any $\tilde{x}^{(0)}$, Newton's*

method (Definition 0.13) gives $\tilde{x}^{(1)} = \mathbf{S}^{-1}\tilde{c}$, the unique minimizer of f .

i Proof

Proof. Partial derivatives of a sum or of a constant multiple are the sum or multiple of the partial derivatives, so by [derivative of a quadratic form](#) for $\frac{1}{2}\tilde{x}^\top \mathbf{S}\tilde{x}$ and [derivative of a linear combination](#) for $\tilde{c}^\top \tilde{x}$, $\frac{\partial}{\partial \tilde{x}} f(\tilde{x}) = \frac{1}{2}(2\mathbf{S}\tilde{x}) - \tilde{c} = \mathbf{S}\tilde{x} - \tilde{c}$; by [Hessian of a quadratic form](#) with $\frac{1}{2}\mathbf{S}$ in place of $\mathbf{S}$, and because the gradient $-\tilde{c}$ of the linear term is constant ([constant with respect to a vector](#)), $\mathbf{H}_f(\tilde{x}) = \mathbf{S}$, which is invertible ([positive definite inverse](#)). So

$$\begin{aligned}\tilde{x}^{(1)} &= \tilde{x}^{(0)} - \mathbf{S}^{-1}(\mathbf{S}\tilde{x}^{(0)} - \tilde{c}) && \text{(definition of Newton's method)} \\ &= \tilde{x}^{(0)} - \mathbf{S}^{-1}\mathbf{S}\tilde{x}^{(0)} + \mathbf{S}^{-1}\tilde{c} && \text{(distribute)} \\ &= \tilde{x}^{(0)} - \tilde{x}^{(0)} + \mathbf{S}^{-1}\tilde{c} && (\mathbf{S}^{-1}\mathbf{S} = \mathbf{I}_p) \\ &= \mathbf{S}^{-1}\tilde{c}. && \text{(cancel)}\end{aligned}$$

The gradient at $\mathbf{S}^{-1}\tilde{c}$ is $\mathbf{S}\mathbf{S}^{-1}\tilde{c} - \tilde{c} = \tilde{0}_{p \times 1}$, and f is strictly convex (Theorem 0.10, part 2), so $\mathbf{S}^{-1}\tilde{c}$ is a global minimizer (Corollary 0.1). It is the only one: a strictly convex function has at most one global minimizer, since the midpoint of two distinct global minimizers, which have equal values, would have a strictly smaller value (Definition 0.5 with $t = \frac{1}{2}$). $\square$

Example 0.38 (One step to the minimizer). Let $f(\tilde{x}) = \frac{1}{2}(x_1^2 + 3x_2^2)$, so $\mathbf{S} = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}$ and $\tilde{c} = \tilde{0}_{2 \times 1}$. From $\tilde{x}^{(0)} = (3, 1)^\top$ the gradient is $\mathbf{S}\tilde{x}^{(0)} = (3, 3)^\top$, and $\mathbf{S}^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1/3 \end{bmatrix}$, so the Newton step is $-\mathbf{S}^{-1}(3, 3)^\top = -(3, 1)^\top$ and $\tilde{x}^{(1)} = (3, 1)^\top - (3, 1)^\top = \tilde{0}_{2 \times 1}$, the minimizer.

Definition 0.16 (Gradient descent). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first partial derivatives, let $\tilde{x}^{(0)}$ be a starting point, and let $s^{(0)}, s^{(1)}, \dots$ be positive **step lengths**. **Gradient descent** computes, for $t = 0, 1, 2, \dots$,

$$\tilde{x}^{(t+1)} \stackrel{\text{def}}{=} \tilde{x}^{(t)} - s^{(t)} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)}).$$

It is Newton's method (Definition 0.13) with the Hessian replaced by $\frac{1}{s^{(t)}}\mathbf{I}_p$, so it needs no second derivatives.

Example 0.39 (Fixed steps on a parabola). For $f(x) = x^2$, $f'(x) = 2x$, so gradient descent with fixed step length s is $x^{(t+1)} = x^{(t)} - 2s x^{(t)} = (1 - 2s)x^{(t)}$, and $x^{(t)} =$

$$(1 - 2s)^t x^{(0)}.$$

- With $s = 0.25$, $x^{(t)} = 0.5^t x^{(0)}$, which tends to the minimizer 0.
- With $s = 0.5$, the first step lands on 0, since $1 - 2s = 0$.
- With $s = 1.5$, $x^{(t)} = (-2)^t x^{(0)}$, which moves away from 0 for any $x^{(0)} \neq 0$: too long a step can make gradient descent fail.

Definition 0.17 (Line search and exact line search). In gradient descent (Definition 0.16), a **line search** is a rule that chooses the step length $s^{(t)}$ by examining the values of f along the line $\{\tilde{x}^{(t)} - s \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)}) : s > 0\}$. **Exact line search** is the line search that chooses $s^{(t)}$ to minimize $f(\tilde{x}^{(t)} - s \frac{\partial}{\partial \tilde{x}} f(\tilde{x}^{(t)}))$ over $s > 0$.

Example 0.40 (Exact line search on a parabola). For $f(x) = x^2$ and $x^{(t)} = 2$, the derivative is $f'(2) = 4$, so the line is $\{2 - 4s : s > 0\}$ and the values along it are

$$\begin{aligned} \phi(s) &= f(2 - 4s) \quad (\text{the value at step length } s) \\ &= (2 - 4s)^2 \quad (f(x) = x^2) \end{aligned}$$

$\phi(s) \geq 0$ for every s , and $\phi(0.5) = (2 - 2)^2 = 0$, so exact line search chooses $s^{(t)} = 0.5$, which lands on the minimizer $x^{(t+1)} = 0$ in one step.

Definition 0.18 (Descent direction). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have a gradient $\tilde{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} f(\tilde{x})$ at $\tilde{x}$. A direction $\tilde{d}$ (direction) is a **descent direction** for f at $\tilde{x}$ if $\tilde{g}^\top \tilde{d} < 0$. When $\tilde{g} \neq \tilde{0}_{p \times 1}$, $-\tilde{g}$ is a descent direction, since $\tilde{g}^\top (-\tilde{g}) = -\|\tilde{g}\|^2 < 0$.

Example 0.41 (Descent directions for $x_1^2 + x_2^2$ at $(1, 0)^\top$). Let $f(\tilde{x}) = x_1^2 + x_2^2$ and $\tilde{x} = (1, 0)^\top$, so $\tilde{g} = (2x_1, 2x_2)^\top = (2, 0)^\top$.

- $\tilde{d} = (-1, 1)^\top$ is a descent direction: $\tilde{g}^\top \tilde{d} = -2 + 0 = -2 < 0$.
- $\tilde{d} = (0, 1)^\top$ is not: $\tilde{g}^\top \tilde{d} = 0$.
- $-\tilde{g} = (-2, 0)^\top$ is: $\tilde{g}^\top (-\tilde{g}) = -4 < 0$.

Theorem 0.14 (A short enough step downhill decreases the function). Let $f : \mathbb{R}^p \rightarrow \mathbb{R}$ have first partial derivatives that are continuous on $\mathbb{R}^p$ (continuity), and let $\tilde{x} \in \mathbb{R}^p$ be a point with $\tilde{g} \stackrel{\text{def}}{=} \frac{\partial}{\partial \tilde{x}} f(\tilde{x}) \neq \tilde{0}_{p \times 1}$. Then there is an $\bar{s} > 0$ with $f(\tilde{x} - s\tilde{g}) < f(\tilde{x})$ for every $s \in (0, \bar{s})$: a short enough step in the descent direction $-\tilde{g}$ (Definition 0.18) decreases f .

i Proof

Proof. Let $h(s) \stackrel{\text{def}}{=} f(\tilde{x} - s\tilde{g})$. The inner function $s \mapsto \tilde{x} - s\tilde{g}$ has derivative $-\tilde{g}^\top$ ([vector-valued derivative](#)), so by the vector chain rule ([vector chain rule](#))

$$\begin{aligned} h'(s) &= -\tilde{g}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{x} - s\tilde{g}), & (\text{chain rule}) \\ h'(0) &= -\tilde{g}^\top \frac{\partial}{\partial \tilde{x}} f(\tilde{x}) & (\text{set } s = 0) \\ &= -\tilde{g}^\top \tilde{g} & (\text{definition of } \tilde{g}) \\ &= -\|\tilde{g}\|^2 & (\text{squared length}) \\ &< 0. & (\tilde{g} \neq \tilde{0}_{p \times 1}) \end{aligned}$$

Since $h'(0) = \lim_{s \rightarrow 0} (h(s) - h(0))/s$ is negative, taking $\varepsilon = |h'(0)|$ in the definition of the limit gives an $\bar{s} > 0$ such that $(h(s) - h(0))/s < h'(0) + |h'(0)| = 0$ for $0 < s < \bar{s}$. Multiplying by $s > 0$, $h(s) < h(0)$, that is, $f(\tilde{x} - s\tilde{g}) < f(\tilde{x})$. $\square$

Example 0.42 (How short is short enough). In Example 0.39, $f(x) = x^2$ at $x \neq 0$ has $g = 2x$, and $f(x - 2sx) = (1 - 2s)^2 x^2 < x^2$ exactly when $|1 - 2s| < 1$, that is, when $0 < s < 1$. So Theorem 0.14 holds with $\bar{s} = 1$, and the step $s = 1.5$ of that example was too long.

Example 0.43 (Gradient descent zigzags). Let $f(\tilde{x}) = \frac{1}{2}(x_1^2 + bx_2^2)$ with $b > 0$, so $\frac{\partial}{\partial \tilde{x}} f(\tilde{x}) = (x_1, bx_2)^\top$ and one gradient step of length s from $\tilde{x}$ gives $((1-s)x_1, (1-bs)x_2)^\top$. For exact line search (Definition 0.17), let $\phi(s) \stackrel{\text{def}}{=} f$ at that point:

$$\begin{aligned} \phi(s) &= \frac{1}{2}((1-s)^2 x_1^2 + b(1-bs)^2 x_2^2), & (\text{definition of } f) \\ \phi'(s) &= -(1-s)x_1^2 - b^2(1-bs)x_2^2 & (\text{chain rule, term by term}) \\ &= -x_1^2 + sx_1^2 - b^2x_2^2 + b^3sx_2^2 & (\text{expand the products}) \\ &= s(x_1^2 + b^3x_2^2) - (x_1^2 + b^2x_2^2). & (\text{collect the terms in } s) \end{aligned}$$

For $\tilde{x} \neq \tilde{0}_{2 \times 1}$, $x_1^2 + b^3x_2^2 > 0$, so ϕ' is increasing in s with a single zero, and ϕ is minimized at $s^* = \frac{x_1^2 + b^2x_2^2}{x_1^2 + b^3x_2^2}$. From $\tilde{x}^{(0)} = (b, 1)^\top$:

$$\begin{aligned} s^{(0)} &= \frac{b^2 + b^2}{b^2 + b^3} & (\text{substitute } x_1 = b, x_2 = 1) \\ &= \frac{2}{1+b}, & (\text{cancel } b^2) \end{aligned}$$

and, with $r \stackrel{\text{def}}{=} \frac{b-1}{b+1}$,

$$1 - s^{(0)} = \frac{1 + b - 2}{1 + b} = r, \quad (\text{common denominator})$$

$$1 - b s^{(0)} = \frac{1 + b - 2b}{1 + b} = -r, \quad (\text{common denominator})$$

$$\tilde{x}^{(1)} = (r b, -r \cdot 1)^\top = r(b, -1)^\top. \quad (\text{the gradient step from } (b, 1)^\top)$$

For $b \neq 1$ the same holds at every step, by [induction](#) (when $b = 1$, $r = 0$ and $\tilde{x}^{(1)}$ is already the minimizer): if $\tilde{x}^{(t)} = (c b, \pm c)^\top$ for a number $c \neq 0$, then s^* , which depends only on $x_1^2 = c^2 b^2$ and $x_2^2 = c^2$, is again $\frac{2}{1+b}$ (the factor c^2 cancels), and the step multiplies the first entry by r and the second by $-r$. So

$$\tilde{x}^{(t)} = (b r^t, (-r)^t)^\top, \quad f(\tilde{x}^{(t)}) = \frac{1}{2}(b^2 + b) r^{2t} = r^{2t} f(\tilde{x}^{(0)}).$$

For $b \neq 1$, exactly one entry changes sign at every step: the second when $b > 1$ (then $r > 0$), and the first when $b < 1$ (then $r < 0$). So the iterates zigzag toward the minimizer $\tilde{0}_{2 \times 1}$.

- With $b = 3$, $r = \frac{1}{2}$: $\tilde{x}^{(1)} = (1.5, -0.5)^\top$ and $\tilde{x}^{(2)} = (0.75, 0.25)^\top$, and f falls from 6 to 1.5 to 0.375, a quarter at each step.
- With $b = 1$, $r = 0$ and one step reaches the minimizer.
- With $b = 0.01$, $r^2 = (0.99/1.01)^2 \approx 0.96$, so f shrinks by only about 4% per step. Newton's method reaches the minimizer of any such f in one step (Theorem [0.13](#)).

Further reading

- Boyd and Vandenberghe (2004) is the standard reference on convex optimization, covering convex sets, convex functions, convex optimization problems, duality, unconstrained minimization, and interior-point methods.
- Nocedal and Wright (2006) is a comprehensive reference on continuous numerical optimization, covering line search methods, trust-region methods, conjugate gradient methods, quasi-Newton methods, and nonlinear equations.
- [Hua Zhou's lecture notes for "UCLA Biostat 216 - Mathematical Methods for Biostatistics" \(2024 Fall\)](#)
- [Neural Networks](#) is a YouTube playlist by Grant Sanderson (3Blue1Brown); its chapters on gradient descent and backpropagation calculus visually illustrate how gradients of multivariate cost functions are computed and used for optimization.

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